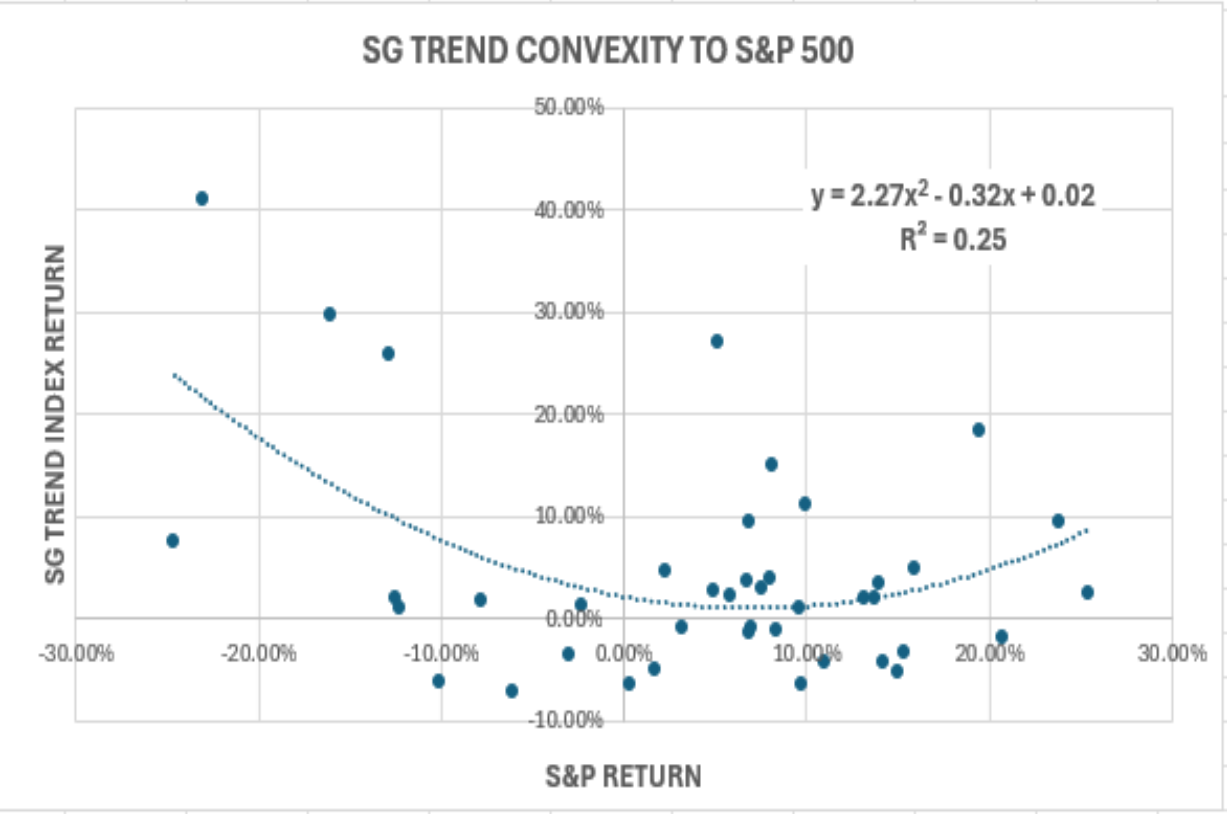
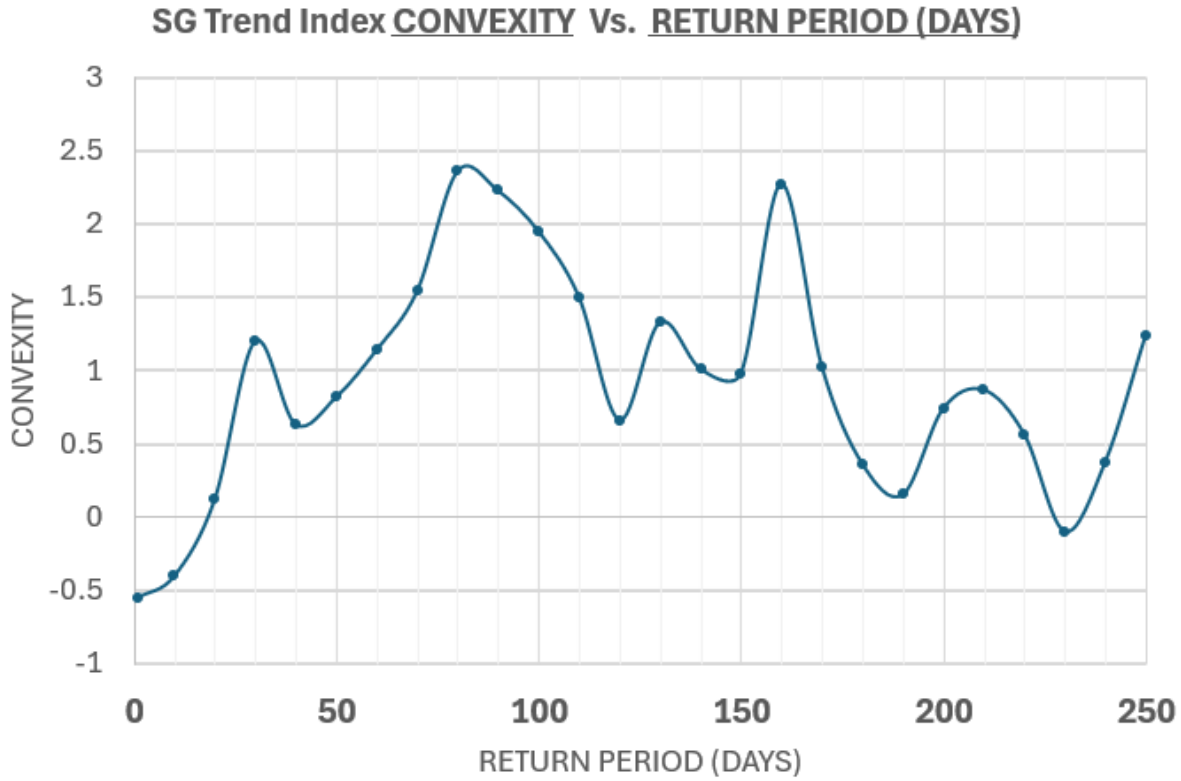
SG TREND INDEX CONVEXITY VS S&P 500

I defined the measure of convexity as the coefficient of the square term in a quadratic line-of-best fit.



I plotted the convexity against the return period in days (from 1 to 250 days) to examine which return periods correspond to maximum values of the convexity term



SG TREND INDEX CONVEXITY VS S&P 500

Key takeaways

- Convexity peaks at approximately an 80-90 day return period.
- The spike at 160 days is driven by a single outlier (the 160 days prior to 16/3/2001).
- Convexity generally quickly improve as the return period extends from daily to roughly 80-90 days, before declining again.
- This seems consistent with Robertson, Goodall/Man AHL (2023) and highlights the importance of choosing the time frame over which to assess the effectiveness of Trend Following strategies.
- The 90-day period seems long enough for trend following strategies to identify a trend, yet too long to respond to any sudden, short-term reversals.

Future Work

- Obtain a convexity measure beyond a simple quadratic line-of-best-fit using other convex curves.
- The impact that risk control strategies have on explaining the convexity in addition to the length of the trend signal. Factors such as reducing position size, cutting total risk, and asset selection are likely as important, as mentioned by **Tom Wrobel** in recent industry discussions.